

Rajat Masanagi

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Education

Bachelor of Technology, Mumbai University 2022 – 2026 | Mumbai
Computer Science and Engineering (Data Science), Honours in Computational Finance CGPA - 9.43/10.0

Delhi Public School 2020 – 2022 | Navi Mumbai
12th CBSE-95%

Podar International School 2020 | Navi Mumbai
10th ICSE-96.6%

College Coursework

Relevant Coursework: Operating Systems, Database Systems, Probability and Statistics, Data Structures and Algorithms, Communication Networks, Machine Learning, Deep Learning, Reinforcement Learning, Data Engineering, Big Data, Artificial Intelligence, Image Processing and Computer Vision, Time Series Analysis, Geospatial Data Science, Social Network Analysis, Natural Language Processing, Recommender Systems, Large Language Models, Information Security, High Performance Computing, Data Ethics, Web Engineering, Economic Modeling, Quantitative Portfolio Management, Financial Analysis and Decision Science, Financial Market and Computing

Experience

Trainee Analyst, Wissen Technology 07/26 – Present | Mumbai
• Completing Training Courses in Java, Springboot and AI Frameworks.

SDE Intern, Swiggy 09/2025 – 06/2026 | Bengaluru
• Automated operational SQL workflows using Python and PostgreSQL via a conversational LLM interface built with LangChain. Supported 10-message context, chat history. Enabled secure prompt-to-SQL execution with sanitization and guardrails, direct result rendering, and LangSmith observability with Feedback loops. Built an automatic visualization of suggestions for the query results using Groq to return structured chart metadata (**Won AI Explorer Award Feb'26**).
• Designed a high-performance spreadsheet alternative for large datasets using React, AG Grid, FastAPI, and PostgreSQL, supporting lazy-loaded server-side querying, parallel bulk updates, background exports, n8n-integrated email delivery, virtual columns, VLOOKUP, pivot tables, and audit logs; reduced analysis time by **8 man-hours per day** across **500k+ row datasets** and **improved FnV fill rate by up to 2%**, directly contributing to FnV GMV.
• Built an in-house ZPL-based label-printing system with a Python UI, delivering an intuitive and user-friendly UX for warehouse staff while **saving ₹26L annually**.
• Prototyped OCR and vision-based solutions for extracting manufacturing and expiry dates, built n8n + Python workflows for concurrent warehouse picking and picker-assignment queries, developed an IoT pipeline using Tapo devices for real-time temperature/humidity monitoring and alerts, assisted in migrating dashboards from Retool to ToolJet, and enhanced the Picking Algorithm to support multiple putaway locations and single-batch pod assignment.

Software Development Intern, Dimension Six Technologies 02/2025 – 07/2025 | Mumbai
• Designed a real-time rear-view vehicle detection and risk assessment system for cyclists using YOLOv8 and a custom COCO-filtered dataset, with risk scoring based on proximity, velocity, acceleration, optical flow, and lane invasion. Developed a Flutter-based mobile interface with UDP integration to display detected threats and trigger real-time voice alerts in dangerous situations.
• Optimized the inference pipeline by converting models to Edge TPU-TPU-compatible TensorFlow Lite, achieving a **95.5% reduction in inference time** and enabling low-cost deployment on Raspberry Pi 4 with Google Coral TPU (**32% cheaper** than M.2 Hailo).

AI-ML Intern, Ultraceuticals 11/2024 – 04/2025
• Fetched drug-target interaction data from a MySQL database and identified potential drug candidates with docking scores less than -7 for specific targets using UniProt IDs.
• Scraped and processed research papers related to identified drugs and targets, and built a Retrieval-Augmented Generation (RAG) system using Gemini to build agents to assist researchers in discovering novel drug-target relationships.
• Applied a Missing Link Graph Neural Network (GNN) to predict new drug-target interactions.

Projects

Scalable Event Booking Platform 🌐, | Java, Spring Boot, Angular, MySQL, Redis, Docker, Nginx, JMeter 08/2026
• Developed a scalable event-ticketing platform using Java and Spring Boot microservices, implementing JWT authentication, role-based access control, Redis-based queueing and caching, concurrent ticket reservations, expiring inventory holds, idempotent checkout, and MySQL persistence.
• Load-tested the Dockerized platform using JMeter with 1,000 concurrent users, processing 7,000 requests at 143 requests/second with 0% errors and 100% checkout completion for successfully admitted users.

Lunar Surface Navigation System, (1st Runner Up in PS ISRO BAH '25) 06/2025
• Analyzed Chandrayaan-2 data to identify safe navigation paths on the Moon's South Pole, aligning datasets through georeferencing and detecting terrain hazards using techniques like YOLOv8, Digital Terrain Models (DTM), and Hough Transform with Canny Edge Detection, achieving optimal results with the latter.
• Evaluated and compared over 5+ path planning algorithms, selecting PRM-RL for its superior performance. Identified potential water ice sites using Imaging Infrared Spectrometer data (3 μm hydration feature) and developed an interactive 3D rover simulation in Three.js to visualize the planned path.

Crop Recommendation Engine (IIIT Lucknow HackoFiesta 6.0 Grand Finalist)

09/2024

- Developed an interactive geospatial dashboard using ISRO’s Bhuvan data to visualize soil parameters such as organic content, surface moisture, evapotranspiration, and soil depth, along with current weather, 5+ years of trends, and 6+ month LSTM forecasts.
- Built an end-to-end crop recommendation pipeline with OCR-based N-P-K extraction from soil reports, price forecasting for 300+ crops based on harvesting timelines and APMC proximity, and a Gemini Agent integrating GIS, weather, soil, and market data for recommendations.

Smart Waste Management System (Winner at PICT Techfiesta International Hackathon '24)

04/2024

- Developed a waste-management platform that detects bin fill levels using OpenCV contour analysis and routes trucks only to full bins using the Capacitated Vehicle Routing Problem (CVRP) algorithm with Google Maps API. Included features like barcode-based certificate generation and YOLOv8-powered litter detection via CCTV and facial detection.
- Built intelligent assistance tools for waste-disposal queries and upcycling recommendations using Gemini, with a full-stack implementation in Node.js, Flutter, and Django.

Adaptive Quiz EdTech Platform (2nd Runner Up in Datahack 3.0)

10/2024

- Developed an AI learning platform using React.js, Flask, Node.js, Firebase, LLaMA, LLaVA, Gemini API, and RAG to generate quizzes, flashcards, answer document-based questions, and read diagrams. Implemented Q-Learning (RL) to personalize question difficulty based on user performance.
- Built a management portal for flashcards, companies, and employees with TTS-based video generation for accessible learning content.

No-Code Workflow Generator (Media.net aiVolution Hackathon '25 Finalist)

10/2025

- Built a full-stack AI workflow automation platform using React.js, FastAPI, ReactFlow, and Agno, converting natural language prompts into executable visual workflows with dynamic workflow graph generation and multi-agent orchestration.
- Designed an asynchronous execution engine with concurrent task scheduling, modular node composition, persistent workflow storage, and seamless integration with OpenAI, Gemini, and Groq for model-agnostic workflow execution.

Healthcare CRM, (built at Mumbai Hacks '25)

11/2025

- Built dual portals for hospitals and doctors using Node.js, Express.js, TypeScript, PostgreSQL, and Prisma, featuring a 3D AI Avatar receptionist for natural language Re-Act SQL querying, with 55+ REST endpoints enforcing capacity, billing, and automated patient transfer workflows.
- Developed real-time dashboards with patient management, severity-tiered alerts, inventory low-stock detection, and a live environmental feed (weather, AQI, city events) for regional demand forecasting.

Geospatial Data Analysis for Tourism

06/2025

- Built a geospatial analytics framework using ResNet-50 feature extraction on geo-tagged Flickr images to analyze tourism patterns and identify points of interest across Mumbai.
- Applied novel Hierarchical Gaussian Latent Process Variable Model (HGLPVM) on ResNet-50 extracted features to identify hidden gem locations within high-density sub-clusters.

Automatic Ad Optimization

02/2025

- Built an AI-powered ad optimization platform using Agentic AI, LLMs, and Vertex AI to automate budget allocation, ad generation, personalization, and performance tracking.
- Developed AI agents to analyze user behavior, optimize targeting, and integrated WhatsApp and Twitter to improve organic reach and campaign engagement.

Publications

Hybrid Quantum-Classical Brain Tumor Segmentation

09/2026

International Journal on Smart Sensing and Intelligent Systems (IJSSIS)

- Replaced the CNN bottleneck of a 3D U-Net with a simulated 6-qubit VQC using PennyLane, angle encoding, RX/RZ rotations, and CNOT entanglement, achieving Dice scores WT: 0.89, TC: 0.82, ET: 0.78 on BraTS 2020 with only ~36 trainable quantum parameters.
- Conducted a 9-model ablation study across Classical, Attention, and Quantum U-Net variants, achieving ~21% ECE calibration improvement via Monte Carlo Dropout and PennyLane-based quantum variance estimation.

Skills

Programming Languages: Python, Java, C, MySQL
Machine Learning: TensorFlow, PyTorch, Scikit-Learn, Numpy, Pandas, RL, Time Series, Recommender Systems, OpenCV, NLP
AI / GenAI: LLMs, LangChain, Groq, LangGraph, Agentic AI, RAG
Web Development & APIs: FastAPI, Flask, Streamlit, Springboot HTML, CSS, REST APIs
Databases and tools: PostgreSQL, MongoDB, ChromaDB, Git, Gephi, QGIS, Tableau

Achievements

7x Hackathon Winner 
Winner at Sanshodhan 2.0, Hack To Crack 2.0, Err_404 6.0, Datahack 3.0, NEXUS ML Competition, Techfiesta 1.0, Technograd 1.0.
ISRO Bharatiya Antariksh Hackathon | Top 30 / 34,000 participants
Bajaj Finserv Health HackRX 6.0 | Top 11 / 45,000+ participants
PanIIT Alumni Association IMAGINE Hackathon | Top 72 / 15,000 participants